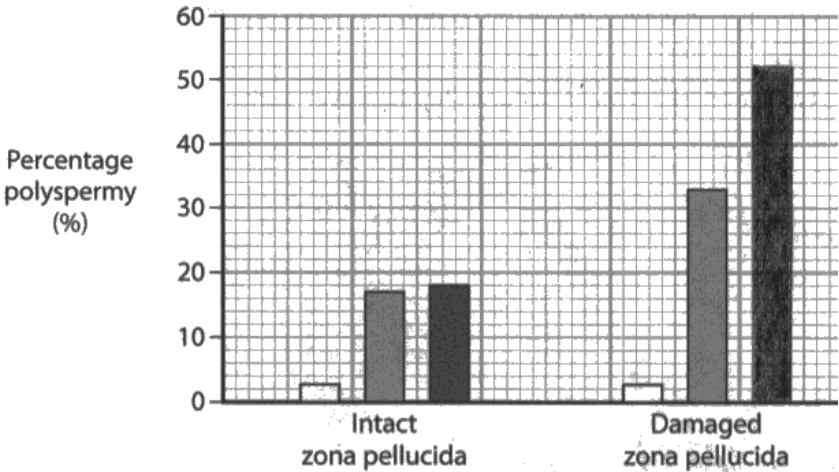


Question Number	Answer	Additional guidance	Mark
8(a)	An answer that makes reference to the following points: - packaged into a (secretory) vesicle by the Golgi {apparatus / body} (1) (vesicle) fuses with cell surface membrane to release them outside of the cell (1)	ACCEPT transported from the Golgi {apparatus / body} in (secretory) vesicles ACCEPT exocytosis ACCEPT exocytosis in wrong context e.g. acrosomes	(2)

Question Number	Answer	Additional guidance	Mark
8(b)	An answer that makes reference to the following points: - individual sperm have slower speeds than bundled sperm (in all resistances) (1) - increase in resistance (above 0.5) decreased the speed of individual sperm more than bundled sperm / individual sperm speed decreases with increased resistance whereas bundled sperm speed {increases / stays relatively constant} (1) (because) {bundles of sperm / more flagella} can generate more force (1) {lower confidence / no significant difference} in results for 0.5 resistance due to overlap of {SD / error / range} bars / resistance has little effect on the speed of the sperm bundle as the error bars for each of the resistance overlap (1)	ACCEPT converse ACCEPT {largest / quoted comparative} difference in speed at 4.0 resistance / bundled sperm have double the speed at 4.0 resistance ACCEPT converse ACCEPT converse for other viscosities	(4)

Question Number	Answer	Additional guidance	Mark												
8(c)(i)	correctly plotted bar with shading (1)	 A bar chart titled 'Percentage polyspermy (%)' on the y-axis, which ranges from 0 to 60 in increments of 10. The x-axis has two categories: 'Intact zona pellucida' and 'Damaged zona pellucida'. Each category has three bars: a white bar, a grey bar, and a black bar. For 'Intact zona pellucida', the values are approximately 2%, 17%, and 18% respectively. For 'Damaged zona pellucida', the values are approximately 2%, 33%, and 52% respectively. <table border="1"> <thead> <tr> <th>Zona Pellucida Status</th> <th>White Bar (%)</th> <th>Grey Bar (%)</th> <th>Black Bar (%)</th> </tr> </thead> <tbody> <tr> <td>Intact</td> <td>~2</td> <td>~17</td> <td>~18</td> </tr> <tr> <td>Damaged</td> <td>~2</td> <td>~33</td> <td>~52</td> </tr> </tbody> </table>	Zona Pellucida Status	White Bar (%)	Grey Bar (%)	Black Bar (%)	Intact	~2	~17	~18	Damaged	~2	~33	~52	(1)
Zona Pellucida Status	White Bar (%)	Grey Bar (%)	Black Bar (%)												
Intact	~2	~17	~18												
Damaged	~2	~33	~52												

Question Number	Answer	Additional guidance	Mark
8(c)(ii)	An explanation that makes reference to four of the following points: • there was more polyspermy of egg cells when the zona pellucida was damaged / converse for intact zona pellucida (1) • due to cortical reactions only occurring over some part of the egg cell (1) • (therefore some) areas of the egg cell not being covered by a hardened zona pellucida (1) • there was more polyspermy of egg cells with a higher sperm concentration • increased sperm cell concentration increased the likelihood of sperm cells entering {egg cell before cortical reaction occurred / through gap in hardened zona pellucida} (1)	ACCEPT {fewer/no} cortical reactions (with damaged zona pellucida) / converse for intact zona pellucida ACCEPT (therefore damaged) zona pellucida does not {harden / thicken} / converse for intact zona pellucida ACCEPT increased sperm cell concentration increased the {concentration of digestive enzymes / rate of digestion of zona pellucida} ignore increased {concentration / damaged to zona pellucida} increases number of sperm reaching the egg cell	(4)